

Understanding Accessibility Sufficiency, Activity Participation, and Quality of Life: Evidence from Austin, TX

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INTRODUCTION

Motivation: Current research and practice lacks guidance on how benefits and burdens from accessibility can be distributed in the fairest way, by offering moral reasoning to guide us towards a fairer distribution.

Previous studies on accessibility and transport equity have often focused on disparities in accessibility without addressing the normative question of what constitutes a sufficient level of access for marginalized populations

Aims and Objectives: This study aims to investigate how transportation accessibility influences residents' daily activity participation and overall quality of life in the city.

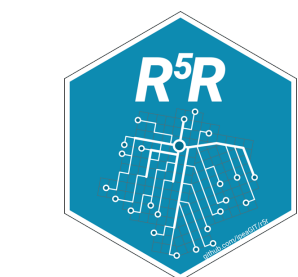
- Visualizing accessibility and transit accessibility varies for destinations
- Quantifying the relationship between accessibility and non-work activity participation using negative binomial activity-generation models
- Describing the relationship between accessibility, activity participation and quality of life.

DATA & METHODS

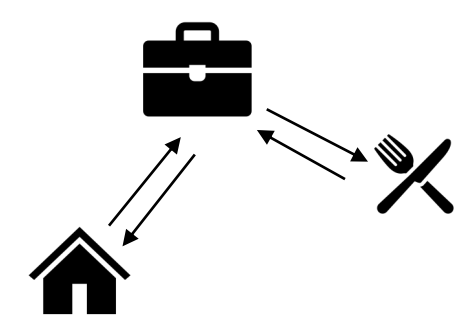
Data Sources: HIFLD Geo-platform for POIs, 2017 Capital Area MPO Household Travel Survey data for activities/trips calculation, and 2018 City of Austin Community Survey data to measure quality of life. CBG-level demographic data from American Community Survey (ACS) 2016-2020,

Accessibility Measure: We measured accessibility non-work points of interests (POIs) –grocery stores and supermarkets, hospitals, urgent cares, colleges and universities, and parks from population-weighted centroids of census block groups (CBGs). We computed accessibility in R using the *R5R* package, with input from OpenStreetMap and General Transit Feed Specification (GTFS) data for Austin for a weekday morning peak period (7am-9am).

Activity Participation: We calculated activity participation as number of daily activities per person without double counting activities of the same purpose which occur at the same location, following the method used by Allen & Farber (2020).



Accessibility
R5R functions used
accessibility(),
travel_time_matrix()
cost_to_closest()



Activity Participation:
Total Trips: 4
Total Nonwork Activities: 1
Unique Non-Work Activities: 1

Quality of Life: Please rate your satisfaction with the following.	
1. The City of Austin as a place to live	
2. The City of Austin as a place to work	
3. The City of Austin as a place to raise children	
4. The City of Austin as a place to retire	
5. The City of Austin as a place where I feel welcome	
6. Overall quality of life in the city	

Quality of Life measure:
Respondents' satisfaction rating on a scale of 1-5 have been used to calculate average QoL measure

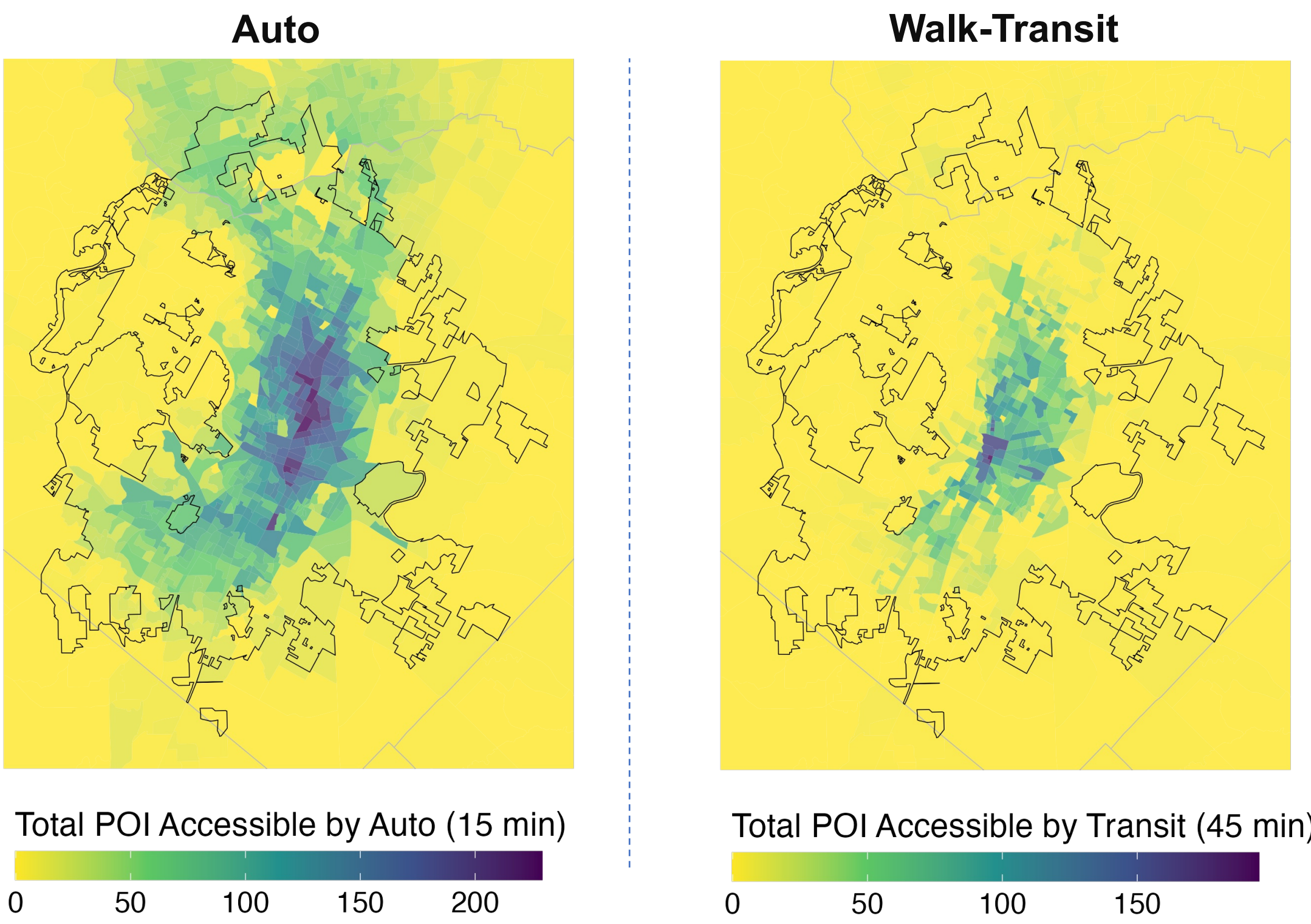
Primal Accessibility: Counting how many opportunities can be reached in a given travel time (e.g., 15 min of auto or 45 min of transit).

Dual Accessibility: Measuring the time required to reach N closest destinations of a certain type.

RESULTS

Primal Accessibility

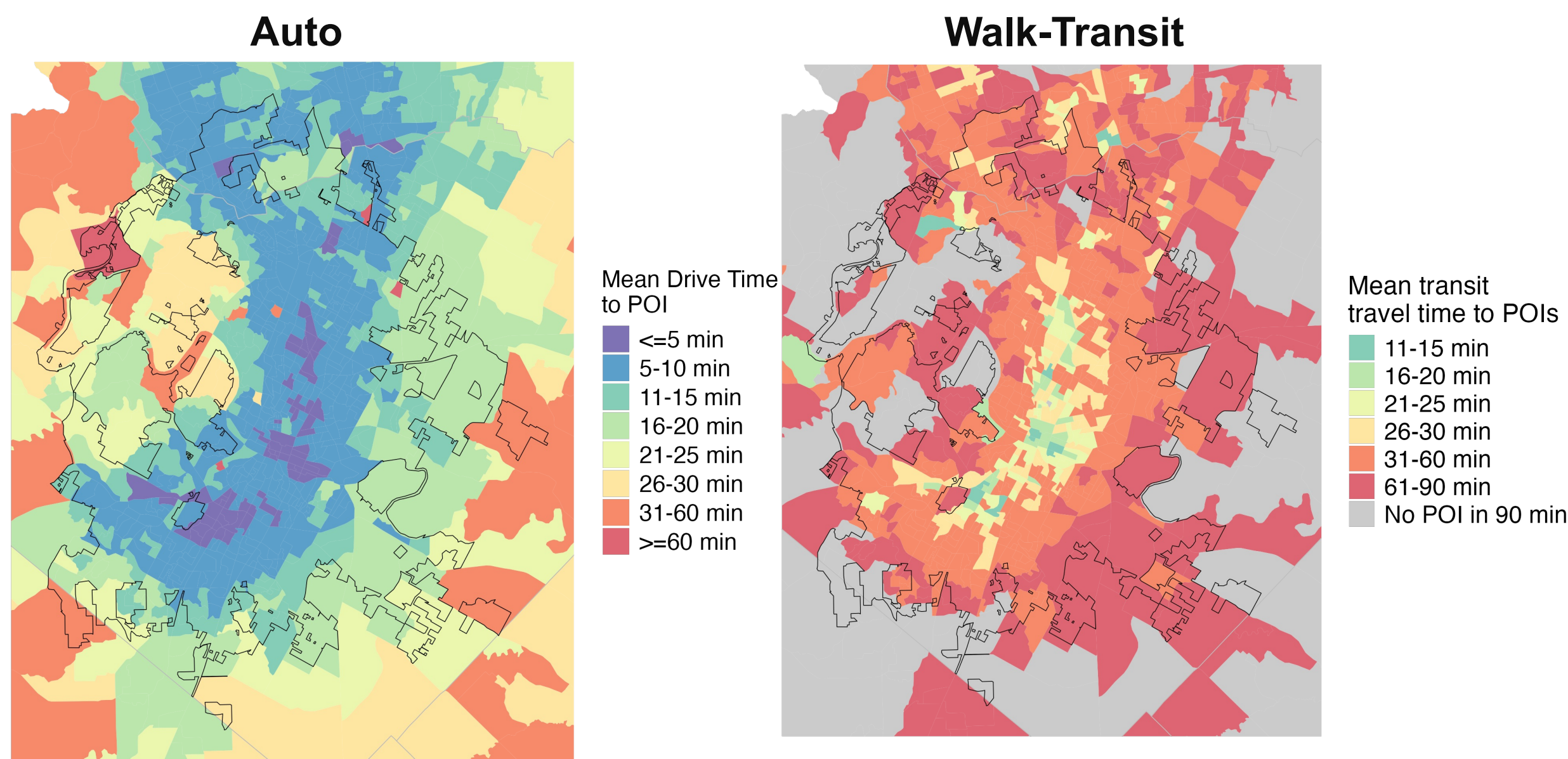
Total POIs accessible by auto and transit



Auto offering nearly 11 times more essential destinations over a shorter time frame.

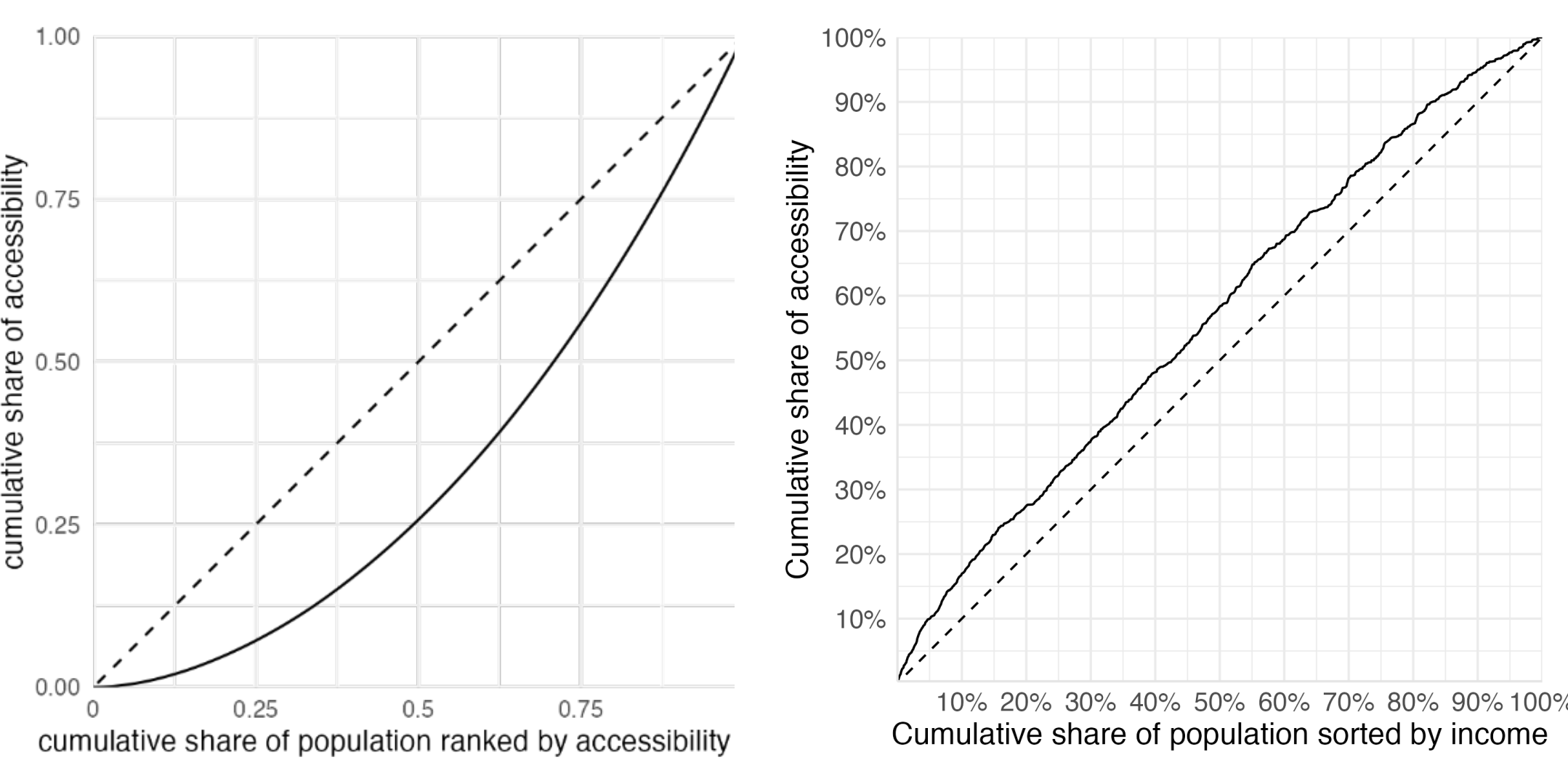
Dual Accessibility

Average auto and transit travel time to reach nearest POIs



The average travel time to the nearest POIs is 15 minutes by car and 42 minutes by transit, meaning transit takes nearly 2.75 times longer than driving on average.

Accessibility Inequality



The Palma Ratio (0.671) and Concentration Index (-0.045) both indicate that the accessibility distribution slightly favors the poorer population.

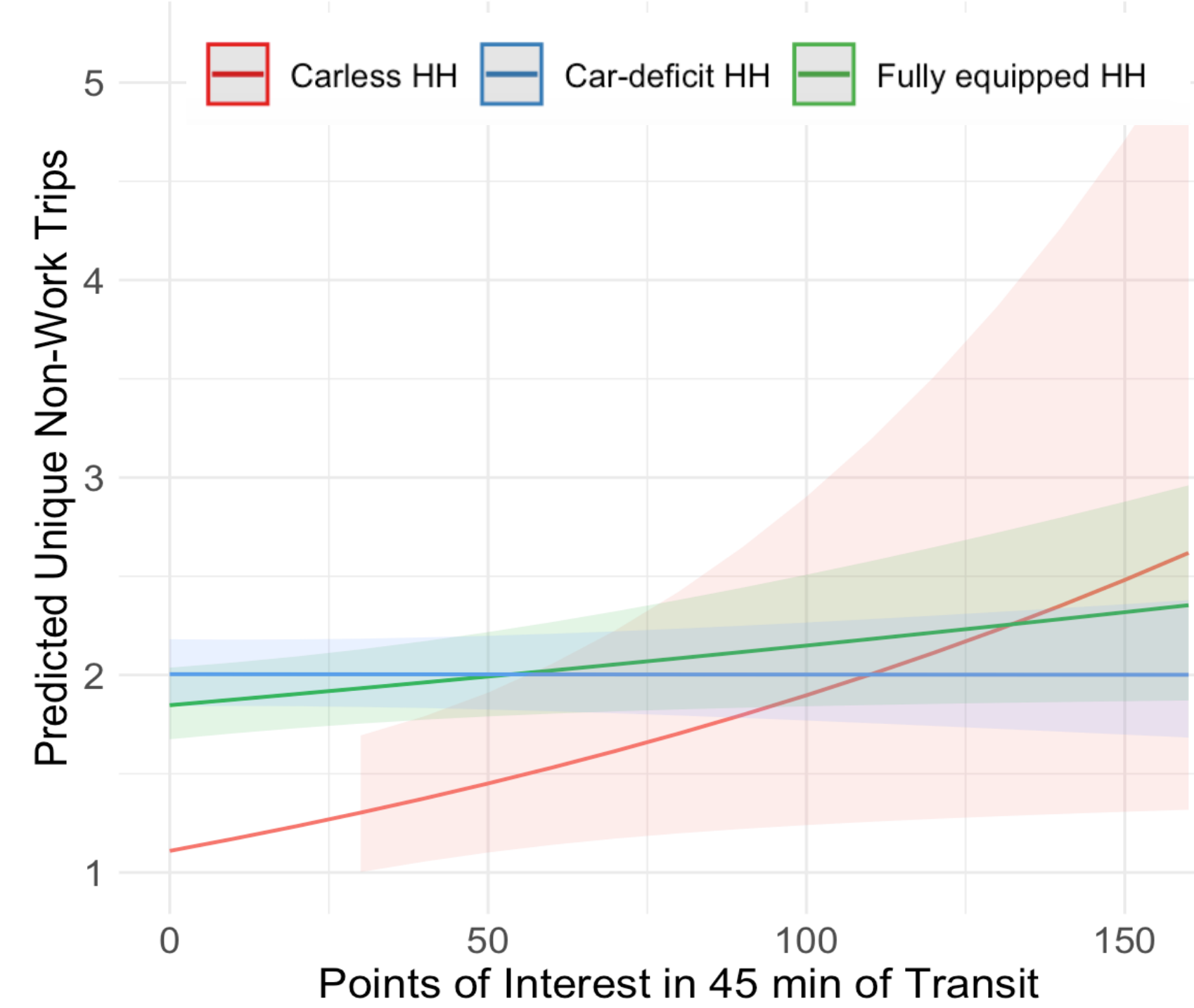
Accessibility-Activity Participation Model

Independent Variables	Dependent variables: [Activity participation]			
	Total Non-Work Activity Trips	IRR	Unique Non-Work Activity Trips	IRR
Accessibility Measure				
Transit Primal Accessibility (45 min Walk-Transit)	1.0014	0.0014***	1.0012	0.0012***
Auto Dual Accessibility (Average of travel times to nearest POIs) Inversed	0.9945	-0.0055***	0.9948	-0.0052***
Personal Characteristics				
Sex (ref Female)				
Male	0.8641	-0.1460***	0.8672	-0.1424***
Age (ref 31-65)				
18-30	0.7978	-0.2259***	0.8795	-0.1284***
66-75	0.9776	-0.0226	0.9967	-0.0033
75+	0.8156	-0.2038***	0.8725	-0.1364***
Race (ref White)				
Black	1.0082	0.0081	0.9613	-0.0394
Hispanic/Latino	1.0304	0.0299	0.9941	-0.0059
Asian/Pacific Islander	0.8514	-0.1609***	0.8525***	-0.1595***
Native American	0.9190	-0.0844	0.9433	-0.0584
Other	0.9554	-0.0457	0.9472	-0.0542
Student Status (ref Not a student)				
Student	1.6842	0.5213***	1.5683	0.4500***
Disability (No)				
Yes	0.8844	-0.1228**	0.8653	-0.1446**
Flex Time (ref Fixed/Unchanging)				
Flexible	1.1165	0.1102***	0.8653	0.0386
Not Employed	0.8804	-0.1274***	0.8518	-0.1604***
Household Characteristics				
Household Income (ref [\$50,000-\$99,999])				
[< \$50,000]	1.0376	0.0370	1.0319	-0.0314
[\$150,000 or more]	1.0310	0.0305	1.0478	0.0153
Number of Worker	0.9194	-0.0841***	0.9352	-0.0710***
Household with Children (ref no children)				
With children	1.2691	0.2383***	1.2088	0.1897***
Vehicle Ownership Status (ref Car Deficit (0 < VA < 1))				
Zero Car (VA=0)	0.9235	-0.0795*	0.8579*	-0.1533
Fully equipped household (VA > =1)	1.0805	0.0774**	1.0457**	0.0345
Household Bike (ref Zero bike)				
1 or more	1.0697	0.0674***	1.0437	0.0428*
Neighborhood Selection (ref. Non access factors)				
Access related factors	1.0419	0.0410*	1.0372	0.0365
Residence Type (ref Single Family)				
Multi-family Housing	1.0604	0.0587*	1.0358	0.0352
Neighborhood Characteristics				
Activity Density (ref low density)				
Medium density	0.8869	-0.1200**	0.9042	-0.1007**
High density	0.9302	-0.0723*	0.9408	-0.0610*
(Intercept)	1.0543	0.0529***	1.4611	0.3792***
Observations	5725		5725	
R ² Nagelkerke	0.152		0.133	
AIC	21634.01		18028.17	

(*p ≤ 0.1, **p ≤ 0.05, ***p ≤ 0.01)

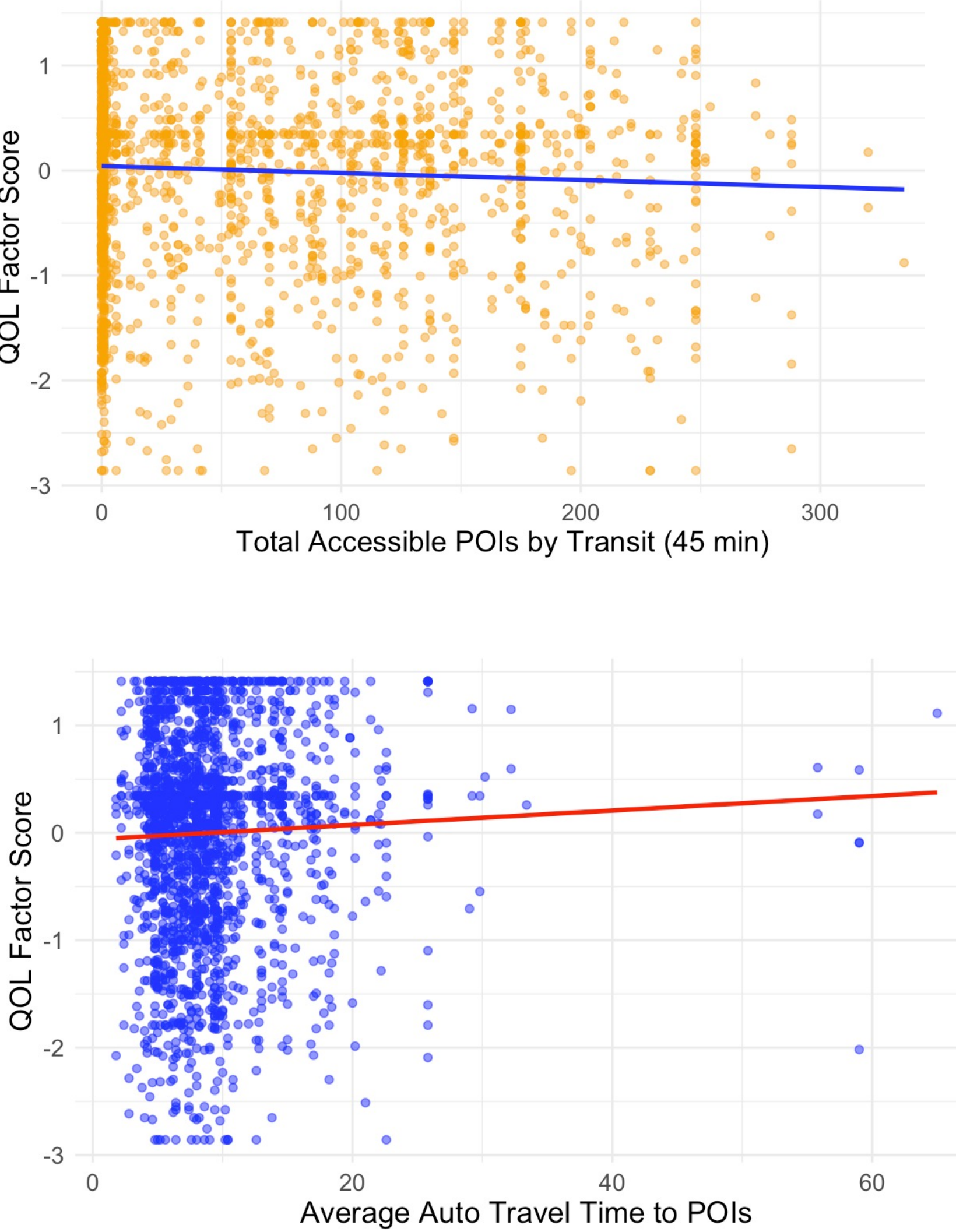
Both auto and transit accessibility is associated with a small but significant increase in both total non-work activity trips and unique non-work activity trips. Activity trips increase slightly with enhanced accessibility, suggesting that current levels are insufficient to meet the needs of low-income, carless, or car-deficit households.

Stratified Model



Transit Accessibility plays a critical role for carless households, enabling higher participation in non-work activities.

ACCESSIBILITY & QOL



Although weak relationships are observed, better auto accessibility (shorter average travel times to POIs) and better transit accessibility (higher total accessible POIs by transit) are associated with lower Quality of Life (QOL) scores.

CONCLUSION

Despite residing in areas with relatively better spatial accessibility, low-income, carless households are unable to meet their activity needs under the current level of accessibility, resulting in a persistently low quality of life.

Solely measuring activity participation may not reliably reflect the experienced benefits from spatial accessibility, as it does not indicate the extent to which desired destinations can be reached

Hence sufficient accessibility level determination requires perceived accessibility measure of the individual.

Research need to consider both measured and perceived accessibility to define sufficient level of access for individual or household.

FUNDING



NATIONAL CENTER FOR UNDERSTANDING FUTURE
TRAVEL BEHAVIOR AND DEMAND (TBD)

National University Transportation Center sponsored by US
Department of Transportation